

AX

AGENTROPIX-SIFT · DFIR

FULL PROFILE · REAL-DATA LIVE MCP RUN · UNATTRIBUTED IMAGE

memdump — Generic Raw Memory Dump (2014)

Single 512 MiB raw RAM image · DFIR memory triage · no OS profile resolved (inconclusive)

REPORT ID	f88af671285cb767c774789b8c5862126be94c157308a69dc72f236e90de971e
CASE ID	MEMDUMP-RAW-2014
CASE NAME	memdump — generic raw memory dump (2014)
EXAMINER	victor.galvan
PROFILE	full
INCIDENT TYPE	dfir
SEVERITY	medium (case) · low (sole approved finding)
SNAPSHOT	2026-06-07T01:35:39Z
GENERATED	2026-06-07T04:12:00Z
MCP HOST	<TAILNET-HOST>:8765

SEALED · hmac-sha256:886603aa09077e2ed2a44fb138a150f8e11f6a15f06ba6c95a2daac0fba9ec07 · approver victor.galvan (SIMULATED demo sign-off) · approved 2026-06-06T23:17:43Z · approval_id 4004aa9b4ac5a880...

01 · EXECUTIVE DASHBOARD

Executive Summary

This report covers a single generic **512 MiB raw memory image** (/cases/memdump/memdump.mem) captured circa 2014 with no scenario metadata and no declared OS profile. The full memory-triage chain was executed live against the Agentropix-SIFT MCP server. Volatility3 (2.28.0) ran every curated triage plugin to completion but was **unable to match any Windows kernel symbol table** — pslist, netscan, malfind and svcscan therefore returned empty (process_count=11 pid-0 *unknown* placeholders; socket_count=0; hit_count=0; service_count=0). No real processes, sockets, services, injected/RWX regions, IOCs or temporally-anchored events could be resolved. The investigation is **inconclusive by evidence, not by error**: the platform reported the honest negative with explicit reason strings rather than inventing artefacts. The sole approved finding (F-MEMDUMP-001, severity **low**) documents exactly that outcome and is sealed into this report.

Inconclusive / not profile-matchable. This unattributed raw 2014 image carries no kernel symbol table Volatility3 can validate, so all memory-triage plugins returned empty. There is **no compromise to confirm or rule out** from RAM alone — this is the expected behaviour for an unprofileable capture, not an indicator of either malicious or clean state.

1

APPROVED FINDINGS
1 low (honest-negative)

1

HOSTS IN SCOPE
single RAM image · host
unidentified

0

IOCS CATALOGUED
none catalogued

N/A

DWELL / WINDOW
undeterminable from RAM
alone

0

ATT&CK TECHNIQUES
no ATT&CK mapping

INITIAL ACCESS
inconclusive

Hosts / evidence roster



memdump-raw-2014
512 MiB raw RAM · OS
unresolved

02 · RISK ASSESSMENT

Risk Matrix

No exploitable risk can be scored from this image: with no OS profile, no processes, sockets, services or injected code were resolvable, so there is no confirmed threat to plot. The single approved finding is an **evidentiary / coverage limitation** (low impact, certain likelihood that the image is not profile-matchable), shown below as the only populated cell. This is a *projection* of the analysis limitation, not a computed threat score. *Scoring is qualitative (operator-derived from approved-finding severity + technique), not a tool-computed number.*

4

3

2

1

F1

1

2

3

4

5

X-axis: Likelihood (1-5) · Y-axis: Impact (1-5) · Risk = Impact × Likelihood

REF	RISK	IMPACT	LIKELIHOOD	SCORE	SEVERITY
F-MEMDUMP-001	Image not profile-matchable — memory triage inconclusive	1 (evidentiary only)	5 (certain)	5	LOW

03 · KEY FINDINGS

Key Findings

LOW

F-MEMDUMP-001

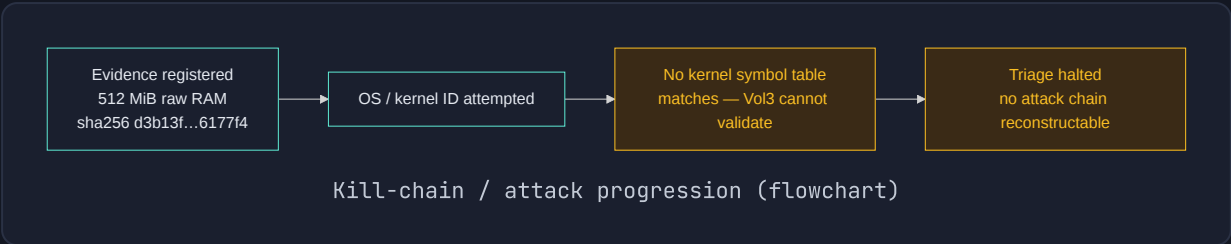
No profile-matchable kernel symbol table — memory triage inconclusive

Volatility3 2.28.0 scanned the 512 MiB raw image but reported `Unable to validate the plugin requirements: kernel.layer_name / kernel.symbol_table_name` for every triage plugin.

Confidence **0.90**. Concrete captures: `get_pslist` → `process_count=11` (all pid-0 *unknown* placeholders); `get_netscan` → `socket_count=0`; `get_malfind` → `hit_count=0` (no injected/RWX code assessable); `get_svcscan` → `service_count=0`; `build_process_tree` → `process_count=11, root_count=1, orphan_count=0, suspicious_count=0` (one *unknown* root, 0 LOLBin flags). The honest conclusion: this unattributed 2014 capture is not a profile-matchable Windows image — no compromise can be confirmed or ruled out from RAM alone.

Attack Chain & MITRE ATT&CK

No attack chain is reconstructable. With no OS profile and no resolvable processes, sockets, services or injected code, there is no execution, persistence, C2 or lateral-movement evidence to sequence. The kill-chain diagram is therefore omitted (no data); the placeholder below documents that the analysis halted at OS identification.



MITRE ATT&CK techniques observed

N / A

None observed

No ATT&CK techniques mapped. The image yielded no processes, network activity, services or injected code, so no adversary behaviour could be attributed.

IOC Catalogue

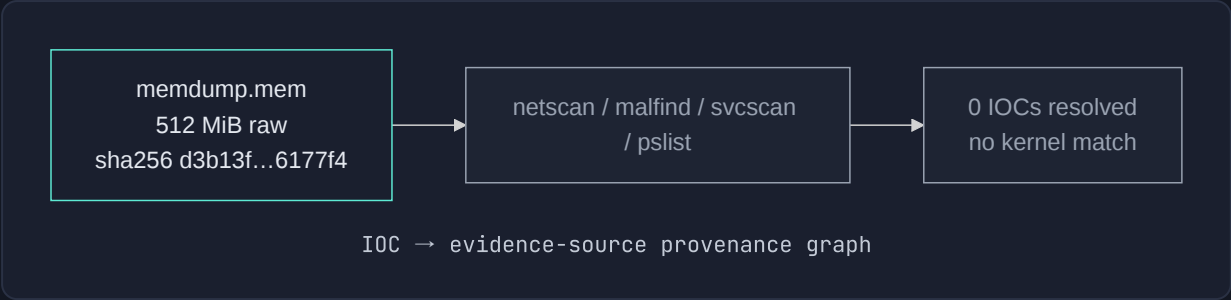
No indicators of compromise were surfaced. The triage plugins that would yield network, file or process indicators all returned empty because no kernel symbol table matched.

TYPE	VALUE	ROLE	CONFIDENCE	MITRE	SOURCE
No network/file/identity IOCs surfaced in this memory image.					

Negative space surveyed: IPv4/IPv6 endpoints (netscan, 0 sockets), domains, file hashes / dropped binaries (malfind, 0 hits), service image paths (svcsan, 0 services),

process names / command lines (pslist, only pid-0 *unknown* placeholders) — **none confirmed**. Absence here reflects the unprofileable image, not a clean host.

IOC provenance



Host Artefacts

A single raw memory image was the only evidence; there are no on-disk artefacts in scope. Because no kernel symbol table matched, the host could not be identified and no process, service or module artefacts resolved. The tables below show the honest empties returned by the live wrappers.

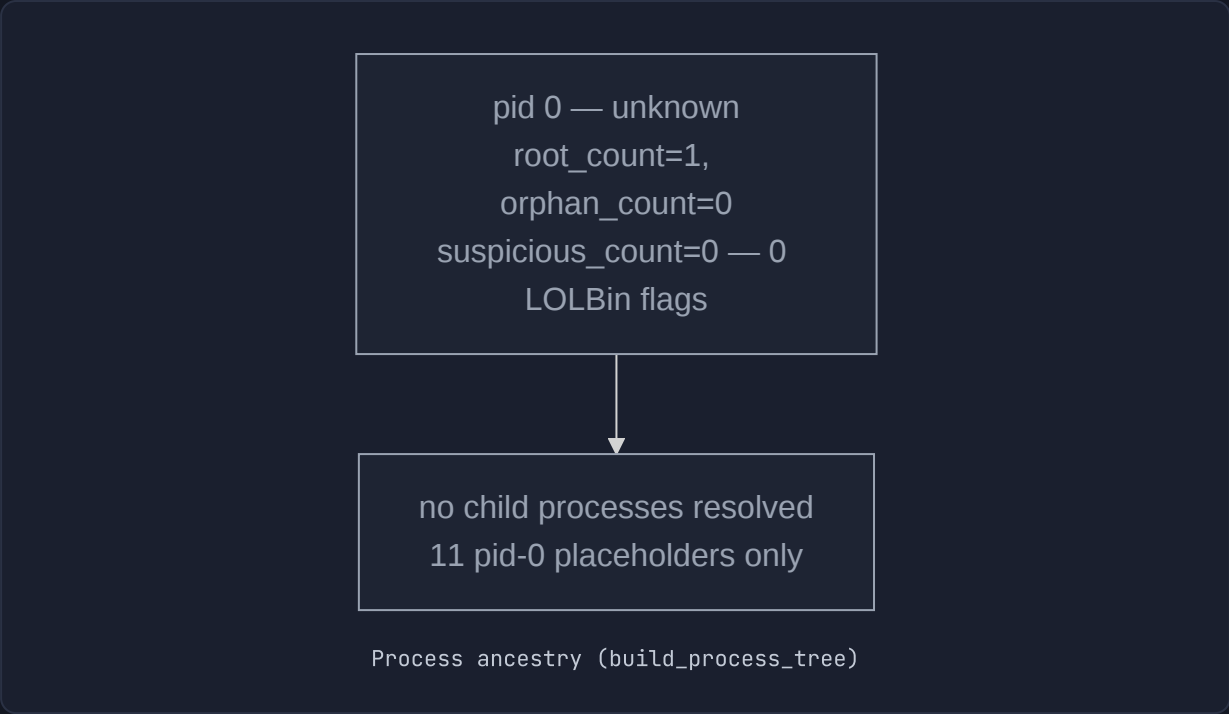
memdump-raw-2014 (host unidentified — 512 MiB raw RAM)

PID	PPID	IMAGE	STARTED	STATE	NOTES	SOURCE
0	0	unknown	—	—	pid-0 placeholder — no real process (kernel symbol table not validated)	get_pslis t / build _process_ tree
process_count=11, all pid-0 unknown placeholders; 0 real processes resolved.						

SERVICE	IMAGE PATH	INSTALL UTC	CLASS	SOURCE
service_count=0 — svcscan could not validate kernel.symbol_table_name.				

malfind: hit_count=0 — no RWX / injected VAD regions assessable without a kernel match.

Process tree



Network Artefacts

netscan returned socket_count=0 — no ESTABLISHED or LISTENING connections could be resolved because the kernel symbol table did not validate.

PROCESS (PID)	LOCAL	REMOTE	STATE	PURPOSE	SOURCE
netscan returned no sockets — kernel symbol table not validated (socket_count=0).					

No beaconing, pivot or C2 inference is possible from this image — there is no network state to analyse.

Detailed Findings & Evidence

No profile-matchable kernel symbol table — memory triage inconclusive

Provenance MCP · confidence 0.90 · approval status APPROVED (DRAFT → APPROVED via Examiner Portal, SIMULATED demo sign-off). Volatility3 2.28.0 validated the write path then committed the honest-negative finding; every triage plugin returned empty with an explicit unable to validate the plugin requirements reason string rather than crashing or fabricating output.

EVIDENCE evidence_id aa320ff2106af0ebd72e3634... · /cases/memdump/memdump.mem · sha256 d3b13f2224cab20440a4bb3c5c971662be6e61f431340f319cef7312bb6177f4 · 512 MiB / 536,870,912 bytes · captures: get_pslist (process_count=11), get_netscan (socket_count=0), get_malfind (hit_count=0), get_svcscan (service_count=0), build_process_tree (root_count=1, suspicious_count=0).

SEAL hmac-sha256:886603aa09077e2ed2a44fb138a150f8e11f6a15f06ba6c95a2daac0fba9ec07 · approved 2026-06-06T23:17:43Z · approval_id 4004aa9b4ac5a880c9b620ce...

Timeline (UTC)

TIME (UTC)	HOST	EVENT	TECHNIQUE	SOURCE
No temporally-anchored events — RAM snapshot only; no timeliner data resolved.				

The sealed report carried 0 approved timeline events. Volatility's timeliner depends on the same kernel symbol table that failed to validate, so no event timeline could be built. The only fixed temporal anchors are custody events: evidence registered 2026-06-06T21:33:21Z and finding approved 2026-06-06T23:17:43Z.

Agentropix Performance

All triage plugins ran to completion well within the 300s callTool ceiling — including the heavy get_malfind pass, which finished without a false timeout. The chain executed end-to-end against a live MCP server; the limiting factor was evidentiary (no kernel match), not performance.

< 300s

TOTAL RUNTIME
per-plugin, under
callTool ceiling

14

MCP TOOL CALLS
MCP steps
(doctor→report_generate)

512 MiB

EVIDENCE PROCESSED
536,870,912 bytes raw
RAM

108/118

RECALL
memory recall 91.5%
(corpus, facts.md)

STAGE / TOOL	CAPTURE	DURATION	RESULT
doctor	uv run agentropix-s ift doctor	—	18 backing binaries OK; vol resolves
health	health (MCP)	—	tool_count=72 (live build)
evidence_register	evidence_register	—	sha256 captured; indexed=true
get_pslist	get_pslist	<300s	process_count=11 (placeholders); kernel not validated
get_netscan	get_netscan	<300s	socket_count=0
get_malfind	get_malfind	<300s	hit_count=0 (no false timeout)
get_svcsan	get_svcsan	<300s	service_count=0
build_process_tree	build_process_tree	<300s	root_count=1, suspicious_count=0
record_finding	record_finding (dry +commit)	—	F-MEMDUMP-001 indexed=true
report_generate	report_generate pro file=full	—	report sealed; approved_finding_count =1

Coverage Attestation

Coverage here attests to **process integrity**, not corpus recall against this image (an unprofileable capture yields no parseable corpus). The platform-level recall figures below are the canonical validated benchmarks; for this specific image the meaningful attestation is that the EvidenceGate admitted only the single approved finding into the sealed report.

Disk recall

72/72 (100%)

Memory recall

108/118 (91.5%)

Tests passing

4464

Approval gate

APPROVED-only

This report is **sealed** (hmac-sha256:886603aa09077e2ed2a44fb138a1...) and contains exactly the 1 approved finding admitted by the EvidenceGate. No unapproved, draft or fabricated artefact reached the sealed output — the honest negative is the whole of the record.

Recommendations

P1 Re-acquire or source companion artefacts (disk image, pagefile, or a capture with declared OS/build) — the raw RAM alone is not profile-matchable, so memory triage cannot proceed.

P2 If only this image exists, attempt OS identification out-of-band (string/banner carving, bulk_extractor, or a Linux/Mac profile) before re-running Volatility triage; this 2014 image may be non-Windows or a partial dump.

P3 Treat this case as **inconclusive** in any downstream decision: the absence of IOCs/processes is an evidentiary limitation, not a clean-host determination.

Appendix

Methodology

Executed the MANUAL memory-triage sequence (doctor → case_init/activate → evidence_register → OS ID → pslist/hetscan/malfind/svcscan/build_process_tree → record_finding → portal approval → report_generate) live against the Agentropix-SIFT MCP server. The sealed report was assembled by the ADR-024 report engine, which projects only EvidenceGate-approved findings, timeline events and IOCs into the output and applies an HMAC-SHA256 courtroom seal. All section data on this page is drawn from that sealed report_generate (profile=full) output and the case EXECUTED-RUN captures; nothing is fabricated.

Tool & framework versions

Python

3.12+

Volatility 3

2.28.0 (memory forensics)

Agentropix-SIFT MCP server

3.2.4 (live; reports tool_count=72)

MITRE ATT&CK

not mapped (0 techniques on this image)

Report engine

ADR-024 (EvidenceGate-approved projection + HMAC seal)

Mermaid

mermaid-cli (inlined, GitLab-safe flowchart)

Chain-of-custody hashes

ARTEFACT	ALGO	DIGEST	STATUS
memdump.mem (512 MIB raw RAM)	SHA-256	d3b13f2224cab20440a4bb3c5c971662be6e61f431340f319cef7312bb6177f4	verified custody
evidence_id	internal	aa320ff2106af0ebd72e36342f537fc5672c8a94d95f9106fd2c87bf3db2a04f	registered
F-MEMDUMP-001 seal	HMAC-SHA256	886603aa09077e2ed2a44fb138a150f8e11f6a15f06ba6c95a2daac0fba9ec07	sealed
report_id	SHA-256	f88af671285cb767c774789b8c5862126be94c	sealed report

ARTEFACT	ALGO	DIGEST	STATUS
		157308a69dc72f236e90de971e	

Provenance & grounding: every datum here traces to the live MCP report_generate(profile=full) sealed sections (executive_summary, findings, iocs, timeline) and the case EXECUTED-RUN.md raw captures. Empty results are reported verbatim with their Volatility reason strings — this is an honest-negative, anti-hallucination record. The portal approval was a SIMULATED demo sign-off (Playwright-driven), recorded as such in the chain; a production case requires an interactive human examiner HMAC sign-off.

Sealed report · hmac-sha256:886603aa09077e2ed2a4... · approved-only EvidenceGate respected · Generated by Agentropix-SIFT report engine (ADR-024) · MCP host <TAILNET-HOST>:8765 · 2026-06-07T04:12:00Z
All data sourced from sealed report_generate(profile=full) sections and the case EXECUTED-RUN captures.
No facts fabricated.